

Cyber Defenders: [Lockdown Lab](#)

Reconstruct a multi-stage intrusion by analyzing network traffic, memory, and malware artifacts using Wireshark, Volatility, and VirusTotal, mapping findings to MITRE ATT&CK.

Scenario: System's SOC has detected suspicious outbound traffic from a public-facing IIS server in its cloud platform activity suggestive of a web-shell drop and covert connections to an unknown host, so we captured the PCAP network traffic file, dumped a full image of the server and pulled a sample malware, we are going to investigate these files to understand how the attack happened and its risks.

Used Tools: Wireshark, Linux Terminal, Strings, Virus Total Website

Investigation Steps:

Investigate PCAP file through Wireshark tool:

Q1: After flooding the IIS host with rapid-fire probes, the attacker reveals their origin. Which IP address generated this reconnaissance traffic?

We have about 6359 packets So, we need to filter this traffic to make it easier to analyze.

The image shows a Wireshark interface with a packet list on the left and a packet details pane on the right. The packet list shows a series of TCP and ARP packets. Packet 6348 is selected, and its details are shown in the pane on the right. The details pane shows the packet structure: Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. The packet data is displayed in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
63...	1380.489...	10.0.2.4	10.0.2.15	TCP	151	4443 → 49688 [PSH, ACK] Seq=2163080 Ack=33213 Win=64128 Len=97
63...	1380.519...	10.0.2.15	10.0.2.4	TCP	54	49688 → 4443 [ACK] Seq=33213 Ack=2163177 Win=2102016 Len=0
63...	1380.548...	10.0.2.15	10.0.2.4	TCP	187	49688 → 4443 [PSH, ACK] Seq=33213 Ack=2163177 Win=2102016 Len=133
63...	1380.579...	10.0.2.15	10.0.2.4	TCP	187	[TCP Retransmission] 49688 → 4443 [PSH, ACK] Seq=33213 Ack=2163177 Win=2102016 Len=133
63...	1380.581...	10.0.2.4	10.0.2.15	TCP	66	4443 → 49688 [ACK] Seq=2163177 Ack=33346 Win=64128 Len=0 SLE=33213 SRE=33346
63...	1440.986...	10.0.2.4	10.0.2.15	TCP	151	4443 → 49688 [PSH, ACK] Seq=2163177 Ack=33346 Win=64128 Len=97
63...	1441.023...	10.0.2.15	10.0.2.4	TCP	54	49688 → 4443 [ACK] Seq=33346 Ack=2163274 Win=2101760 Len=0
63...	1441.034...	10.0.2.15	10.0.2.4	TCP	184	49688 → 4443 [PSH, ACK] Seq=33346 Ack=2163274 Win=2101760 Len=130
63...	1441.036...	10.0.2.4	10.0.2.15	TCP	60	4443 → 49688 [ACK] Seq=2163274 Ack=33476 Win=64128 Len=0
63...	1445.853...	PCSSys...	PCSSys...	ARP	42	Who has 10.0.2.4? Tell 10.0.2.15
63...	1445.854...	PCSSys...	PCSSys...	ARP	60	10.0.2.4 is at 08:00:27:6b:6e:c6
63...	1446.194...	PCSSys...	PCSSys...	ARP	60	Who has 10.0.2.15? Tell 10.0.2.4
63...	1446.194...	PCSSys...	PCSSys...	ARP	42	10.0.2.15 is at 08:00:27:26:b6:da
63...	1501.474...	10.0.2.4	10.0.2.15	TCP	151	4443 → 49688 [PSH, ACK] Seq=2163274 Ack=33476 Win=64128 Len=97
63...	1501.506...	10.0.2.15	10.0.2.4	TCP	54	49688 → 4443 [ACK] Seq=33476 Ack=2163371 Win=2101760 Len=0
63...	1501.526...	10.0.2.15	10.0.2.4	TCP	184	49688 → 4443 [PSH, ACK] Seq=33476 Ack=2163371 Win=2101760 Len=130
63...	1501.527...	10.0.2.4	10.0.2.15	TCP	60	4443 → 49688 [ACK] Seq=2163371 Ack=33606 Win=64128 Len=0
63...	1506.364...	PCSSys...	PCSSys...	ARP	42	Who has 10.0.2.4? Tell 10.0.2.15
63...	1506.365...	PCSSys...	PCSSys...	ARP	60	10.0.2.4 is at 08:00:27:6b:6e:c6
63...	1506.641...	PCSSys...	PCSSys...	ARP	60	Who has 10.0.2.15? Tell 10.0.2.4
63...	1506.641...	PCSSys...	PCSSys...	ARP	42	10.0.2.15 is at 08:00:27:26:b6:da
63...	1512.625...	10.0.2.4	10.0.2.15	TCP	234	4443 → 49688 [PSH, ACK] Seq=2163371 Ack=33606 Win=64128 Len=180
63...	1512.635...	10.0.2.15	10.0.2.4	TCP	54	49688 → 4443 [ACK] Seq=33606 Ack=2163551 Win=2101504 Len=0
63...	1512.684...	10.0.2.15	10.0.2.4	TCP	203	49688 → 4443 [PSH, ACK] Seq=33606 Ack=2163551 Win=2101504 Len=149
63...	1512.686...	10.0.2.4	10.0.2.15	TCP	60	4443 → 49688 [ACK] Seq=2163551 Ack=33755 Win=64128 Len=0

> Frame 6348: Packet, 151 bytes on wire (1208 bits), 151 bytes captured (1208 bits) on interface \DeviceN
> Ethernet II, Src: PCSSystemec_b6:6e:c6 (08:00:27:6b:6e:c6), Dst: PCSSystemec_26:b6:da (08:00:27:26:b6:da)
> Internet Protocol Version 4, Src: 10.0.2.4, Dst: 10.0.2.15
> Transmission Control Protocol, Src Port: 4443, Dst Port: 49688, Seq: 2163274, Ack: 33476, Len: 97
> Data (97 bytes)

0000 08 00 27 26 b6 da 08 00 27 6b 6e c6 08 00 45 00 ...&....'kn...E..
0010 00 89 8c 81 40 00 40 06 95 db 0a 00 02 04 0a 00 ...@_@.....
0020 02 0f 11 5b c2 18 29 10 29 95 2c 02 30 5f 50 18 ...[:):)__0_P..
0030 01 f5 fe 4f 00 00 90 4d c1 aa e3 b9 00 82 f9 6c ...O...M.....l
0040 80 fb 05 79 e5 a9 c1 86 ab d5 90 4d c1 aa 90 4d ...y.....M...M
0050 c1 e3 90 4d c1 aa 90 4d c1 a6 90 4f c1 ab 90 4d ...M...M...O...M
0060 c1 a8 90 4d c1 83 90 4c c1 a8 a6 79 f6 9f a7 7b ...M...L...y...{
0070 f3 9a a8 7d f9 9c a1 7f f6 9b a5 7f f3 9a a4 74 ...:.....t
0080 f5 9d a4 7b f1 99 a3 7d f6 9e 90 4d c1 aa 9c 4d ...{...}...M...M
0090 c3 aa a2 4d c1 aa 90 ...M...

capture.pcapng | Packets: 6359 | Profile: Default

Open Statistics -> Conversations Then we can see the whole connection arranged in talks way.

Conversation Settings		Ethernet - 12 IPv4 - 16 IPv6 - 4 TCP - 1085 UDP - 39													
☐ Name resolution ☐ Absolute start time ☐ Display raw data ☐ Limit to display filter															
Copy Follow Stream... Graph... I/O Graphs															
☐ Protocol ☐ Bluetooth ☐ BPV7 ☐ DCCP ☐ DNP 3.0 ☒ Ethernet ☐ FC ☐ FDDI ☐ IEEE 802.11 ☐ IEEE 802.15.4 ☐ ILNP ☒ IPv4 ☒ IPv6 ☐ IPX ☐ JXTA ☐ LTP ☐ MPTCP ☐ NCP ☐ openSAFETY															
Filter list for specific type															
Address A	Port A	Address B	Port B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A	Flows
10.0.2.4	55475	10.0.2.15	65389	2	114 bytes	320	1	60 bytes	1	54 bytes	78.376228	0.0000			0
10.0.2.4	55475	10.0.2.15	65129	2	114 bytes	566	1	60 bytes	1	54 bytes	78.594639	0.0000			0
10.0.2.4	55475	10.0.2.15	65000	2	114 bytes	756	1	60 bytes	1	54 bytes	78.746170	0.0002			0
10.0.2.4	55475	10.0.2.15	64680	2	114 bytes	248	1	60 bytes	1	54 bytes	78.313203	0.0001			0
10.0.2.4	55475	10.0.2.15	64623	2	114 bytes	913	1	60 bytes	1	54 bytes	78.891648	0.0007			0
10.0.2.4	55475	10.0.2.15	63331	2	114 bytes	737	1	60 bytes	1	54 bytes	78.731128	0.0002			0
10.0.2.4	55475	10.0.2.15	62078	2	114 bytes	997	1	60 bytes	1	54 bytes	78.964699	0.0002			0
10.0.2.4	48298	10.0.2.15	61973	2	128 bytes	1035	1	74 bytes	1	54 bytes	137.903759	0.0000			0
10.0.2.4	55475	10.0.2.15	61900	2	114 bytes	905	1	60 bytes	1	54 bytes	78.885099	0.0000			0
10.0.2.4	55475	10.0.2.15	61532	2	114 bytes	700	1	60 bytes	1	54 bytes	78.701038	0.0001			0
10.0.2.4	55475	10.0.2.15	60443	2	114 bytes	169	1	60 bytes	1	54 bytes	78.239315	0.0000			0
10.0.2.4	55475	10.0.2.15	60020	2	114 bytes	429	1	60 bytes	1	54 bytes	78.475209	0.0003			0
10.0.2.4	55475	10.0.2.15	58080	2	114 bytes	73	1	60 bytes	1	54 bytes	78.144601	0.0000			0
10.0.2.4	55475	10.0.2.15	57797	2	114 bytes	924	1	60 bytes	1	54 bytes	78.901564	0.0002			0
10.0.2.4	55475	10.0.2.15	57294	2	114 bytes	375	1	60 bytes	1	54 bytes	78.425942	0.0000			0
10.0.2.4	55475	10.0.2.15	56738	2	114 bytes	850	1	60 bytes	1	54 bytes	78.831190	0.0002			0
10.0.2.4	55475	10.0.2.15	56737	2	114 bytes	846	1	60 bytes	1	54 bytes	78.828052	0.0001			0
10.0.2.4	55475	10.0.2.15	55600	2	114 bytes	147	1	60 bytes	1	54 bytes	78.220796	0.0000			0
10.0.2.4	55475	10.0.2.15	55555	2	114 bytes	76	1	60 bytes	1	54 bytes	78.149233	0.0000			0
10.0.2.4	55475	10.0.2.15	55056	2	114 bytes	449	1	60 bytes	1	54 bytes	78.492902	0.0000			0
10.0.2.4	55475	10.0.2.15	55055	2	114 bytes	439	1	60 bytes	1	54 bytes	78.483709	0.0000			0
10.0.2.4	55475	10.0.2.15	54328	2	114 bytes	160	1	60 bytes	1	54 bytes	78.231529	0.0005			0
10.0.2.4	55475	10.0.2.15	54045	2	114 bytes	674	1	60 bytes	1	54 bytes	78.679759	0.0000			0
10.0.2.4	55475	10.0.2.15	52869	2	114 bytes	765	1	60 bytes	1	54 bytes	78.755342	0.0001			0
10.0.2.4	55475	10.0.2.15	52848	2	114 bytes	657	1	60 bytes	1	54 bytes	78.665178	0.0002			0
10.0.2.4	55475	10.0.2.15	52822	2	114 bytes	295	1	60 bytes	1	54 bytes	78.351215	0.0002			0
10.0.2.4	55475	10.0.2.15	52673	2	114 bytes	369	1	60 bytes	1	54 bytes	78.419382	0.0001			0
10.0.2.4	55475	10.0.2.15	51493	2	114 bytes	550	1	60 bytes	1	54 bytes	78.583886	0.0002			0
10.0.2.4	55475	10.0.2.15	51103	2	114 bytes	717	1	60 bytes	1	54 bytes	78.715328	0.0006			0
10.0.2.4	55475	10.0.2.15	50800	2	114 bytes	303	1	60 bytes	1	54 bytes	78.358989	0.0000			0
10.0.2.4	55475	10.0.2.15	50636	2	114 bytes	702	1	60 bytes	1	54 bytes	78.703354	0.0000			0
10.0.2.4	55475	10.0.2.15	50500	2	114 bytes	297	1	60 bytes	1	54 bytes	78.353272	0.0000			0
10.0.2.4	55475	10.0.2.15	50389	2	114 bytes	710	1	60 bytes	1	54 bytes	78.710054	0.0000			0
10.0.2.4	55475	10.0.2.15	50300	2	114 bytes	218	1	60 bytes	1	54 bytes	78.282154	0.0000			0
10.0.2.4	55475	10.0.2.15	50006	2	114 bytes	658	1	60 bytes	1	54 bytes	78.666830	0.0000			0
10.0.2.4	55475	10.0.2.15	50003	2	114 bytes	452	1	60 bytes	1	54 bytes	78.496177	0.0003			0
10.0.2.4	55475	10.0.2.15	50002	2	114 bytes	613	1	60 bytes	1	54 bytes	78.632249	0.0000			0
10.0.2.4	55475	10.0.2.15	50001	2	114 bytes	864	1	60 bytes	1	54 bytes	78.846439	0.0004			0

We can see that the attacker flooded our server with packets on nearly all ports in very short time so this is not normal !!!

We got it here, this is his Ip

Ans: 10.0.2.4

Q2: The attacker is carrying out targeted enumeration against the HTTP service on the IIS host. Based on the HTTP request headers, which tool is being used?

to make it easy for us we can filter the search with “http” with the attacker’s IP like that we can detect that the attacker used of the enumeration tools on http “Nmap”, here are some tools like (Gobuster – Nmap – curl – Burp Suite)

No.	Time	Source	Destination	Protocol	Length	Info
2117	94.383711	10.0.2.4	10.0.2.15	HTTP	164	Continuation
2121	94.386424	10.0.2.15	10.0.2.4	HTTP	559	HTTP/1.1 400 Bad Request (text/html)
2131	94.418359	10.0.2.4	10.0.2.15	HTTP	216	PROPFIND / HTTP/1.1
2134	94.421647	10.0.2.4	10.0.2.15	HTTP	201	GET / HTTP/1.1
2140	94.577381	10.0.2.4	10.0.2.15	HTTP	225	GET /lowercheck1725947084 HTTP/1.1
2146	94.888674	10.0.2.4	10.0.2.15	HTTP	205	OPTIONS / HTTP/1.1
2151	94.940677	10.0.2.4	10.0.2.15	HTTP	211	GET /robots.txt HTTP/1.1
2156	94.993777	10.0.2.4	10.0.2.15	HTTP	205	OPTIONS / HTTP/1.1
2160	95.045340	10.0.2.4	10.0.2.15	HTTP	359	POST / HTTP/1.1 (application/x-www-form-urlencoded)
2165	95.148287	10.0.2.4	10.0.2.15	HTTP	263	OPTIONS / HTTP/1.1
2171	95.259995	10.0.2.4	10.0.2.15	HTTP	216	PROPFIND / HTTP/1.1
2175	95.302563	10.0.2.4	10.0.2.15	HTTP	210	GET /git/HEAD HTTP/1.1
2177	99.471619	10.0.2.15	10.0.2.4	HTTP	280	HTTP/1.1 200 OK
2178	99.486134	10.0.2.15	10.0.2.4	HTTP	280	HTTP/1.1 200 OK
2189	99.511851	10.0.2.15	10.0.2.4	HTTP	280	HTTP/1.1 200 OK
2195	99.567332	10.0.2.4	10.0.2.15	HTTP	235	PROPFIND / HTTP/1.1
2196	99.567332	10.0.2.4	10.0.2.15	HTTP	262	OPTIONS / HTTP/1.1
2199	99.570439	10.0.2.15	10.0.2.4	HTTP	280	HTTP/1.1 200 OK
2200	99.589987	10.0.2.15	10.0.2.4	HTTP	1479	HTTP/1.1 404 Not Found (text/html)
2206	99.592760	10.0.2.15	10.0.2.4	HTTP	1570	HTTP/1.1 405 Method Not Allowed (text/html)
2208	99.592802	10.0.2.15	10.0.2.4	HTTP	1570	HTTP/1.1 405 Method Not Allowed (text/html)
2210	99.595131	10.0.2.15	10.0.2.4	HTTP	1570	HTTP/1.1 405 Method Not Allowed (text/html)
2219	99.622022	10.0.2.4	10.0.2.15	HTTP	202	Continuation
2220	99.623318	10.0.2.15	10.0.2.4	HTTP	1570	HTTP/1.1 405 Method Not Allowed (text/html)
2232	99.731226	10.0.2.4	10.0.2.15	HTTP	263	OPTIONS / HTTP/1.1

Ans: nmap

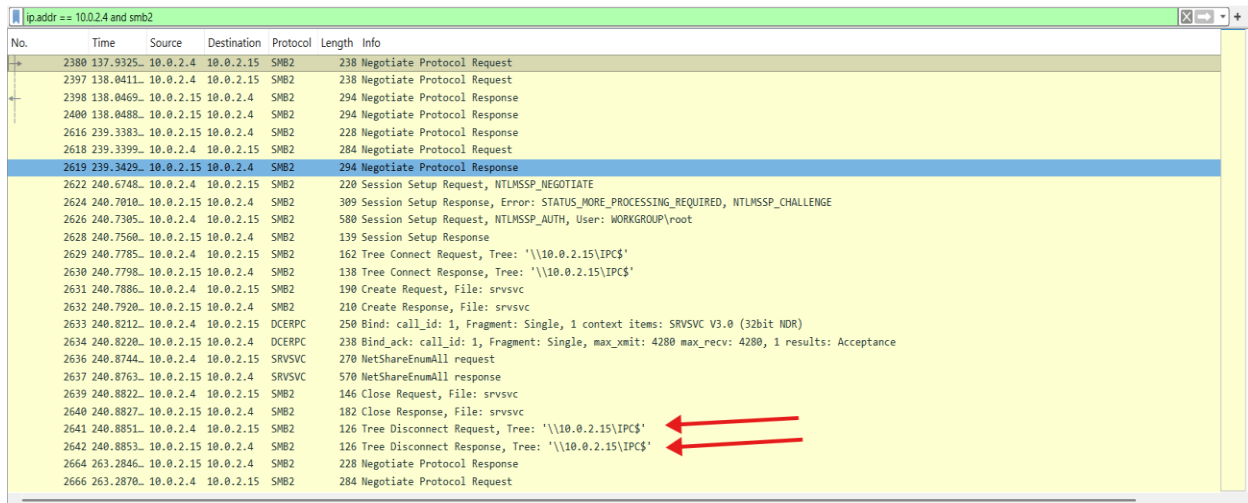
Q3: While reviewing the SMB traffic, you observe two consecutive Tree Connect requests that expose the first shares the intruder probes on the IIS host. Which two full UNC paths are accessed?

We know that SMB Port Is 445 so we can get the paths connected to tree requests with filtration by “SMB” and ip

Hint: there are two types of SMB (just “SMB” for v1 and “SMB2” for the last version)

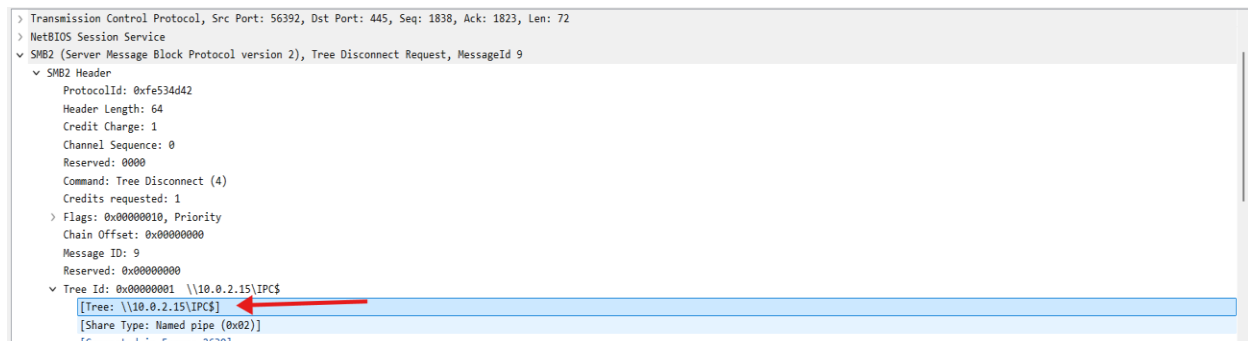
If we search with v1 we will find nothing connected to tree connect

So we can try v2



No.	Time	Source	Destination	Protocol	Length	Info
2380	137.9325..	10.0.2.4	10.0.2.15	SMB2	238	Negotiate Protocol Request
2397	138.0411..	10.0.2.4	10.0.2.15	SMB2	238	Negotiate Protocol Request
2398	138.0469..	10.0.2.15	10.0.2.4	SMB2	294	Negotiate Protocol Response
2400	138.0488..	10.0.2.15	10.0.2.4	SMB2	294	Negotiate Protocol Response
2616	239.3383..	10.0.2.15	10.0.2.4	SMB2	228	Negotiate Protocol Response
2618	239.3399..	10.0.2.4	10.0.2.15	SMB2	284	Negotiate Protocol Request
2619	239.3429..	10.0.2.15	10.0.2.4	SMB2	294	Negotiate Protocol Response
2622	240.6748..	10.0.2.4	10.0.2.15	SMB2	220	Session Setup Request, NTLMSSP_NEGOTIATE
2624	240.7910..	10.0.2.15	10.0.2.4	SMB2	309	Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NTLMSSP_CHALLENGE
2626	240.7905..	10.0.2.4	10.0.2.15	SMB2	580	Session Setup Request, NTLMSSP_AUTH, User: WORKGROUP\root
2628	240.7560..	10.0.2.15	10.0.2.4	SMB2	139	Session Setup Response
2629	240.7785..	10.0.2.4	10.0.2.15	SMB2	162	Tree Connect Request, Tree: '\\10.0.2.15\IPC\$'
2630	240.7798..	10.0.2.15	10.0.2.4	SMB2	138	Tree Connect Response, Tree: '\\10.0.2.15\IPC\$'
2631	240.7886..	10.0.2.4	10.0.2.15	SMB2	190	Create Request, File: srvsvc
2632	240.7920..	10.0.2.15	10.0.2.4	SMB2	210	Create Response, File: srvsvc
2633	240.8212..	10.0.2.4	10.0.2.15	DCERPC	250	Bind: call_id: 1, Fragment: Single, 1 context items: SRVSVC V3.0 (32bit NDR)
2634	240.8220..	10.0.2.15	10.0.2.4	DCERPC	238	Bind_ack: call_id: 1, Fragment: Single, max_xmit: 4280 max_recv: 4280, 1 results: Acceptance
2636	240.8744..	10.0.2.4	10.0.2.15	SRVSVC	270	NetShareEnumAll request
2637	240.8763..	10.0.2.15	10.0.2.4	SRVSVC	570	NetShareEnumAll response
2639	240.8822..	10.0.2.4	10.0.2.15	SMB2	146	Close Request, File: srvsvc
2640	240.8827..	10.0.2.15	10.0.2.4	SMB2	182	Close Response, File: srvsvc
2641	240.8851..	10.0.2.4	10.0.2.15	SMB2	126	Tree Disconnect Request, Tree: '\\10.0.2.15\IPC\$'
2642	240.8853..	10.0.2.15	10.0.2.4	SMB2	126	Tree Disconnect Response, Tree: '\\10.0.2.15\IPC\$'
2664	263.2846..	10.0.2.15	10.0.2.4	SMB2	228	Negotiate Protocol Response
2666	263.2870..	10.0.2.4	10.0.2.15	SMB2	284	Negotiate Protocol Request

Open That Packet and search for a path



```
> Transmission Control Protocol, Src Port: 56392, Dst Port: 445, Seq: 1838, Ack: 1823, Len: 72
> NetBIOS Session Service
  > SMB2 (Server Message Block Protocol version 2), Tree Disconnect Request, MessageId 9
    > SMB2 Header
      ProtocolId: 0xfe534d42
      Header Length: 64
      Credit Charge: 1
      Channel Sequence: 0
      Reserved: 0000
      Command: Tree Disconnect (4)
      Credits requested: 1
    > Flags: 0x00000010, Priority
      Chain Offset: 0x00000000
      Message ID: 9
      Reserved: 0x00000000
    > Tree Id: 0x00000001 \\10.0.2.15\IPC$
      [Tree: \\10.0.2.15\IPC$]
      [Share Type: Named pipe (0x02)]
      [Connected in Name: 2630]
```

If you scroll down with filtration (ip.addr == 10.0.2.4 and smb2)

You can find the next path connected to tree connect requests

ip.addr == 10.0.2.4 and smb2						
No.	Time	Source	Destination	Protocol	Length	Info
2669	263.2951	10.0.2.15	10.0.2.4	SMB2	309	Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NTLMSSP_CHALLENGE
2670	263.2978	10.0.2.4	10.0.2.15	SMB2	580	Session Setup Request, NTLMSSP_AUTH, User: WORKGROUP\root
2671	263.3022	10.0.2.15	10.0.2.4	SMB2	139	Session Setup Response
2672	263.3046	10.0.2.4	10.0.2.15	SMB2	162	Tree Connect Request, Tree: '\\10.0.2.15\IPC\$'
2673	263.3056	10.0.2.15	10.0.2.4	SMB2	138	Tree Connect Response, Tree: '\\10.0.2.15\IPC\$'
2674	263.3069	10.0.2.4	10.0.2.15	SMB2	222	Ioctl Request FSCTL_DFES_GET_REFERRALS, Path: \\10.0.2.15\Documents
2675	263.3076	10.0.2.15	10.0.2.4	SMB2	130	Ioctl Response, Error: STATUS_FS_DRIVER_REQUIRED
2676	263.3090	10.0.2.4	10.0.2.15	SMB2	126	Tree Disconnect Request, Tree: '\\10.0.2.15\IPC\$'
2677	263.3092	10.0.2.15	10.0.2.4	SMB2	126	Tree Disconnect Response, Tree: '\\10.0.2.15\IPC\$'
2678	263.3120	10.0.2.4	10.0.2.15	SMB2	172	Tree Connect Request, Tree: '\\10.0.2.15\Documents'
2679	263.3167	10.0.2.15	10.0.2.4	SMB2	138	Tree Connect Response, Tree: '\\10.0.2.15\Documents'
2681	263.4402	10.0.2.4	10.0.2.15	SMB2	126	KeepAlive Request
2682	263.4409	10.0.2.15	10.0.2.4	SMB2	126	KeepAlive Response
2684	265.8051	10.0.2.4	10.0.2.15	SMB2	179	Create Request, File: <share>
2685	265.8057	10.0.2.15	10.0.2.4	SMB2	210	Create Response, File: <share>
2687	265.8071	10.0.2.4	10.0.2.15	SMB2	156	Find Request, File: <share>, SMB2_FIND_ID_BOTH_DIRECTORY_INFO, Pattern: *
2688	265.8088	10.0.2.15	10.0.2.4	SMB2	614	Find Response, File: <share>, SMB2_FIND_ID_BOTH_DIRECTORY_INFO, Pattern: *, 4 matches
2689	265.8102	10.0.2.4	10.0.2.15	SMB2	156	Find Request, File: <share>, SMB2_FIND_ID_BOTH_DIRECTORY_INFO, Pattern: *
2690	265.8140	10.0.2.15	10.0.2.4	SMB2	130	Find Response, Error: STATUS_NO_MORE_FILES, File: <share>, SMB2_FIND_ID_BOTH_DIRECTORY_INFO, Pattern: *
2691	265.8154	10.0.2.4	10.0.2.15	SMB2	146	Close Request, File: <share>
2692	265.8159	10.0.2.15	10.0.2.4	SMB2	182	Close Response, File: <share>
2693	265.8173	10.0.2.4	10.0.2.15	SMB2	179	Create Request, File: <share>
2694	265.8214	10.0.2.15	10.0.2.4	SMB2	210	Create Response, File: <share>
2695	265.8227	10.0.2.4	10.0.2.15	SMB2	163	GetInfo Request FS_INFO/FileSizeInformation, File: <share>
2696	265.8234	10.0.2.15	10.0.2.4	SMB2	154	GetInfo Response, File: <share>

ANS: \\10.0.2.15\Documents, \\10.0.2.15\IPC\$

Q4: Inside the share, the attacker plants a web-accessible payload that will grant remote code execution. What is the filename of the malicious file they uploaded?

Sharing often uses SMB protocol so we can take a deep look in SMB packets with the attacker's IP

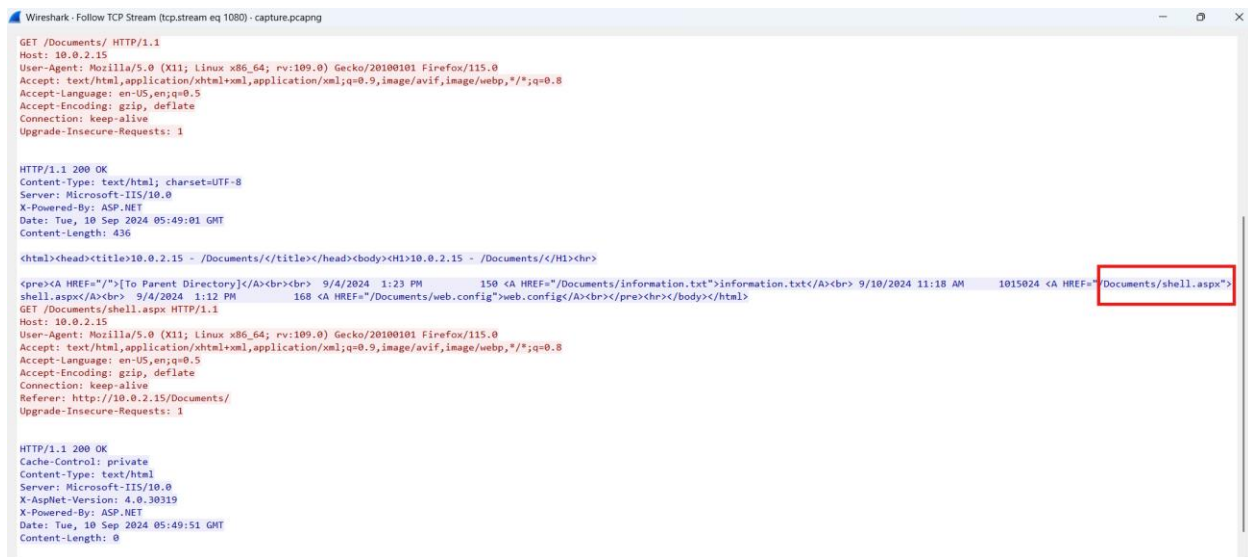
We should find and app that performs commands like ,PowerShell, cmd ,terminal ,etc

ip.addr == 10.0.2.4 and smb2						
No.	Time	Source	Destination	Protocol	Length	Info
2766	322.8890	10.0.2.15	10.0.2.4	SMB2	126	KeepAlive Response
2768	327.8357	10.0.2.4	10.0.2.15	SMB2	126	KeepAlive Request
2769	327.8364	10.0.2.15	10.0.2.4	SMB2	126	KeepAlive Response
2773	332.8947	10.0.2.4	10.0.2.15	SMB2	126	KeepAlive Request
2774	332.8958	10.0.2.15	10.0.2.4	SMB2	126	KeepAlive Response
2778	337.9913	10.0.2.4	10.0.2.15	SMB2	126	KeepAlive Request
2779	337.9920	10.0.2.15	10.0.2.4	SMB2	126	KeepAlive Response
2783	342.4822	10.0.2.4	10.0.2.15	SMB2	198	Create Request, File: shell.aspx
2784	342.5012	10.0.2.15	10.0.2.4	SMB2	210	Create Response, File: shell.aspx
3505	342.6112	10.0.2.4	10.0.2.15	SMB2	494	Write Request Len:1015024 Off:0, File: shell.aspx
3507	342.6124	10.0.2.15	10.0.2.4	SMB2	138	Write Response, File: shell.aspx
3509	342.6142	10.0.2.4	10.0.2.15	SMB2	146	Close Request, File: shell.aspx
3510	342.6160	10.0.2.15	10.0.2.4	SMB2	182	Close Response, File: shell.aspx
3514	342.9192	10.0.2.4	10.0.2.15	SMB2	126	KeepAlive Request
3515	342.9207	10.0.2.15	10.0.2.4	SMB2	126	KeepAlive Response
3517	347.8840	10.0.2.4	10.0.2.15	SMB2	126	KeepAlive Request
3518	347.8853	10.0.2.15	10.0.2.4	SMB2	126	KeepAlive Response
3523	352.8465	10.0.2.4	10.0.2.15	SMB2	126	KeepAlive Request
3524	352.8472	10.0.2.15	10.0.2.4	SMB2	126	KeepAlive Response
3526	357.9074	10.0.2.4	10.0.2.15	SMB2	126	KeepAlive Request
3527	357.9085	10.0.2.15	10.0.2.4	SMB2	126	KeepAlive Response
3531	362.8725	10.0.2.4	10.0.2.15	SMB2	126	KeepAlive Request
3532	362.8728	10.0.2.15	10.0.2.4	SMB2	126	KeepAlive Response
3534	367.9188	10.0.2.4	10.0.2.15	SMB2	126	KeepAlive Request
3535	367.9197	10.0.2.15	10.0.2.4	SMB2	126	KeepAlive Response

Here we find shell.aspx if we followed the http stream with

Right Click -> Follow -> HTTP Stream

We will find the whole packet with shell.aspx and get the first next stream number 1080



```
GET /Documents/ HTTP/1.1
Host: 10.0.2.15
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/115.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate
Connection: keep-alive
Upgrade-Insecure-Requests: 1

HTTP/1.1 200 OK
Content-Type: text/html; charset=UTF-8
Server: Microsoft-IIS/10.0
X-Powered-By: ASP.NET
Date: Tue, 10 Sep 2024 05:49:01 GMT
Content-Length: 436

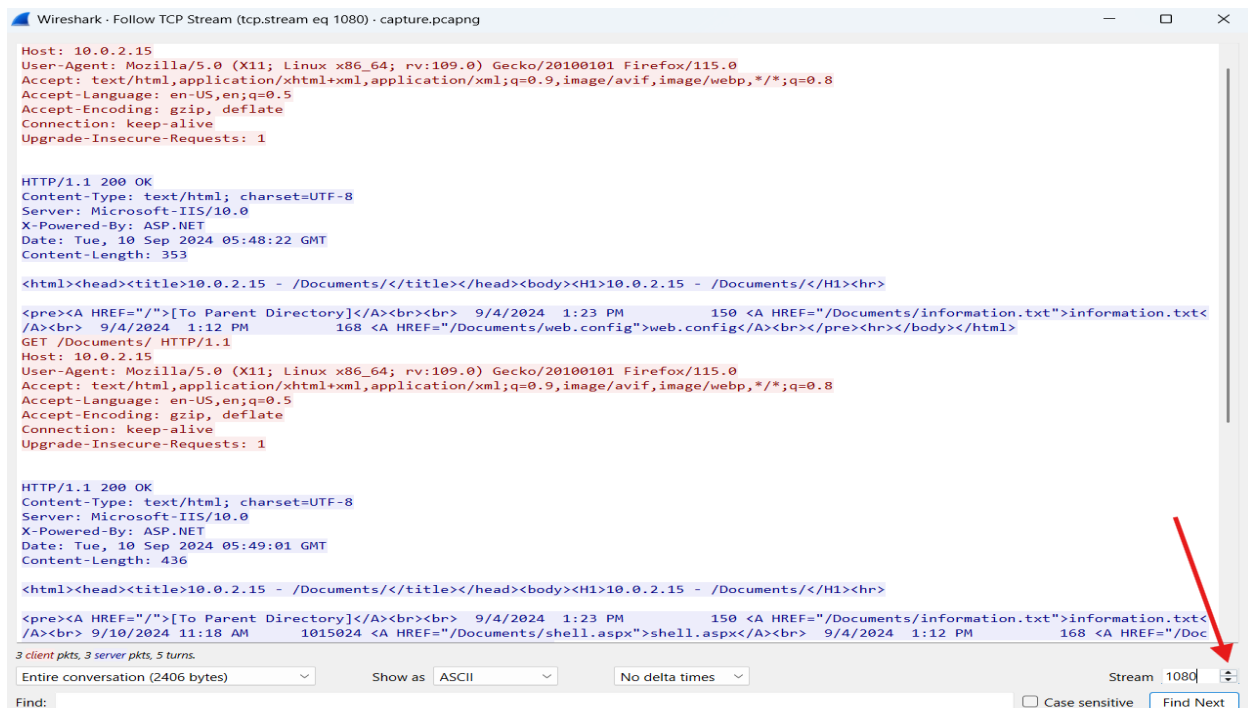
<html><head><title>10.0.2.15 - /Documents/</title></head><body><H1>10.0.2.15 - /Documents/</H1><hr>
<pre><A HREF="/">[To Parent Directory]</A><br><br> 9/4/2024 1:23 PM 150 <A HREF="/Documents/information.txt">information.txt<
/A><br> 9/4/2024 1:12 PM 168 <A HREF="/Documents/web.config">web.config</pre><hr></body></html>
Documents/shell.aspx"
GET /Documents/shell.aspx HTTP/1.1
Host: 10.0.2.15
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/115.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate
Connection: keep-alive
Referer: http://10.0.2.15/Documents/
Upgrade-Insecure-Requests: 1

HTTP/1.1 200 OK
Cache-Control: private
Content-Type: text/html
Server: Microsoft-IIS/10.0
X-AspNet-Version: 4.0.30319
X-Powered-By: ASP.NET
Date: Tue, 10 Sep 2024 05:49:51 GMT
Content-Length: 0
```

ANS: shell.aspx

Q5: The newly planted shell calls back to the attacker over an uncommon but firewall-friendly port. Which listening port did the attacker use for the reverse shell?

After the attacker requests the shell.aspx in stream number 1080 in the first next stream that he would perform his commands in stream 1081



```
Host: 10.0.2.15
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/115.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate
Connection: keep-alive
Upgrade-Insecure-Requests: 1

HTTP/1.1 200 OK
Content-Type: text/html; charset=UTF-8
Server: Microsoft-IIS/10.0
X-Powered-By: ASP.NET
Date: Tue, 10 Sep 2024 05:48:22 GMT
Content-Length: 353

<html><head><title>10.0.2.15 - /Documents/</title></head><body><H1>10.0.2.15 - /Documents/</H1><hr>
<pre><A HREF="/">[To Parent Directory]</A><br><br> 9/4/2024 1:23 PM 150 <A HREF="/Documents/information.txt">information.txt<
/A><br> 9/4/2024 1:12 PM 168 <A HREF="/Documents/web.config">web.config</pre><hr></body></html>
GET /Documents/ HTTP/1.1
Host: 10.0.2.15
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/115.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate
Connection: keep-alive
Upgrade-Insecure-Requests: 1

HTTP/1.1 200 OK
Content-Type: text/html; charset=UTF-8
Server: Microsoft-IIS/10.0
X-Powered-By: ASP.NET
Date: Tue, 10 Sep 2024 05:49:01 GMT
Content-Length: 436

<html><head><title>10.0.2.15 - /Documents/</title></head><body><H1>10.0.2.15 - /Documents/</H1><hr>
<pre><A HREF="/">[To Parent Directory]</A><br><br> 9/4/2024 1:23 PM 150 <A HREF="/Documents/information.txt">information.txt<
/A><br> 9/10/2024 11:18 AM 1015024 <A HREF="/Documents/shell.aspx">shell.aspx</pre><hr></body></html>
1015024 <A HREF="/Documents/shell.aspx">shell.aspx</pre><hr></body></html>
168 <A HREF="/Documents/web.config">web.config</pre><hr></body></html>
3 client pkts, 3 server pkts, 5 turns.
Entire conversation (2406 bytes) Show as ASCII No delta times Stream 1080
Find: Case sensitive Find Next
```

Or we can get the packet with filter (tcp.stream eq 1081)

No.	Time	Source	Destination	Protocol	Length	Info
3585	401.4966	10.0.2.15	10.0.2.4	TCP	66	49688 → 4443 [SYN, ECF, CWR] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
3586	401.4985	10.0.2.4	10.0.2.15	TCP	66	4443 → 49688 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
3587	401.4986	10.0.2.15	10.0.2.4	TCP	54	49688 → 4443 [ACK] Seq=1 Ack=1 Win=2102272 Len=0
3588	401.9507	10.0.2.4	10.0.2.15	TCP	441	4443 → 49688 [PSH, ACK] Seq=1 Ack=1 Win=64256 Len=387
3589	401.9908	10.0.2.15	10.0.2.4	TCP	54	49688 → 4443 [ACK] Seq=1 Ack=388 Win=2101760 Len=0
3590	402.0718	10.0.2.15	10.0.2.4	TCP	175	49688 → 4443 [PSH, ACK] Seq=1 Ack=388 Win=2101760 Len=121
3591	402.0737	10.0.2.4	10.0.2.15	TCP	60	4443 → 49688 [ACK] Seq=388 Ack=122 Win=64256 Len=0
3592	402.1175	10.0.2.4	10.0.2.15	TCP	163	4443 → 49688 [PSH, ACK] Seq=388 Ack=122 Win=64256 Len=109
3593	402.1628	10.0.2.15	10.0.2.4	TCP	54	49688 → 4443 [ACK] Seq=122 Ack=497 Win=2101760 Len=0
3594	402.1638	10.0.2.15	10.0.2.4	TCP	175	49688 → 4443 [PSH, ACK] Seq=122 Ack=497 Win=2101760 Len=121
3595	402.1657	10.0.2.4	10.0.2.15	TCP	60	4443 → 49688 [ACK] Seq=497 Ack=243 Win=64256 Len=0
3596	402.1681	10.0.2.4	10.0.2.15	TCP	163	4443 → 49688 [PSH, ACK] Seq=497 Ack=243 Win=64256 Len=109
3597	402.1812	10.0.2.15	10.0.2.4	TCP	54	49688 → 4443 [ACK] Seq=243 Ack=606 Win=2101760 Len=0
3598	402.2166	10.0.2.15	10.0.2.4	TCP	535	49688 → 4443 [PSH, ACK] Seq=243 Ack=606 Win=2101760 Len=481
3599	402.2538	10.0.2.15	10.0.2.4	TCP	535	[TCP Retransmission] 49688 → 4443 [PSH, ACK] Seq=243 Ack=606 Win=2101760 Len=481
3600	402.2548	10.0.2.4	10.0.2.15	TCP	66	4443 → 49688 [ACK] Seq=606 Ack=724 Win=64128 Len=0 SLE=243 SRE=724
3601	402.7941	10.0.2.4	10.0.2.15	TCP	163	4443 → 49688 [PSH, ACK] Seq=606 Ack=724 Win=64128 Len=109
3602	402.8157	10.0.2.15	10.0.2.4	TCP	54	49688 → 4443 [ACK] Seq=724 Ack=715 Win=2101504 Len=0
3604	402.8454	10.0.2.15	10.0.2.4	TCP	175	49688 → 4443 [PSH, ACK] Seq=724 Ack=715 Win=2101504 Len=121
3605	402.8487	10.0.2.4	10.0.2.15	TCP	60	4443 → 49688 [ACK] Seq=715 Ack=845 Win=64128 Len=0
3608	403.1473	10.0.2.4	10.0.2.15	TCP	1514	4443 → 49688 [ACK] Seq=715 Ack=845 Win=64128 Len=1460
3609	403.1473	10.0.2.4	10.0.2.15	TCP	1514	4443 → 49688 [ACK] Seq=2175 Ack=845 Win=64128 Len=1460
3610	403.1473	10.0.2.4	10.0.2.15	TCP	1514	4443 → 49688 [ACK] Seq=3635 Ack=845 Win=64128 Len=1460
3611	403.1473	10.0.2.4	10.0.2.15	TCP	1514	4443 → 49688 [ACK] Seq=5095 Ack=845 Win=64128 Len=1460
3612	403.1473	10.0.2.4	10.0.2.15	TCP	1514	4443 → 49688 [PSH, ACK] Seq=6555 Ack=845 Win=64128 Len=1460

> Frame 3599: Packet, 535 bytes on wire (4280 bits), 535 bytes captured (4280 bits) on interface \Device...	0000 08 00 27 6b 6e c6 08 00 27 26 b6 da 08 00 45 00 ...km...S&...E...
> Ethernet II, Src: PCSSystemtec_26:b6:da (08:00:27:26:b6:da), Dst: PCSSystemtec_b1:6e:c6 (08:00:27:6b:...	0010 02 09 4d b4 40 00 80 06 00 00 0a 00 02 0f 0a 00 ...H...
> Internet Protocol Version 4, Src: 10.0.2.15, Dst: 10.0.2.4	0020 02 04 c2 18 11 5b 2c 01 ae 8e 28 ef 29 a9 50 18 ...[... ..(.)P...
> Transmission Control Protocol, Src Port: 49688, Dst Port: 4443 Seq: 243, Ack: 606, Len: 481	0030 20 12 1a 0e 00 00 2a b4 ef 55 59 40 2e 7d 43 95 ...*... (Urg.)C...
Source Port: 49688	0040 ae 04 bf 80 cb 56 7b 7f 85 2a 2a b4 ef 55 2a b4 ...V[... ..Y*...T*
Destination Port: 4443	0050 ee 9c 2a b4 ef 54 2a b4 ef 59 2a b6 ef 54 2a b4 ...*...T*...Y*...T*
[Stream Index: 1081]	0060 ef 5f 2a b4 ef 7c 2a b5 ef 57 1f 84 d9 66 13 81 ...*...[... ..W...F...
[Stream Packet Number: 15]	0070 d9 61 13 8c d7 6c 1b 80 d7 6d 1a 80 de 62 1e 84 ...a...1... ..b...
> [Conversation completeness: Incomplete, DATA (15)]	0080 d6 65 19 8d df 6d 13 8c dd 63 2a b4 ef 55 26 b4 ...e...m...e*...U8...
[TCP Segment Len: 481]	0090 ed 55 21 b4 ef 55 3b b4 ef 55 26 b4 ed 55 21 b4 ...U!U!U!U!U!U!U!U!
	00a0 ef 55 39 b4 ef 55 26 b4 ed 55 21 b4 ef 55 38 b4 ...U9-U8-U!-U8-U!
	00b0 ef 55 26 b4 ed 55 21 b4 ef 55 3c b4 ef 55 26 b4 ...U8-U!-U!-U!-U8-U!

ANS:4443

Memory Dump Analysis

Investigation Steps:

Open Kali -> Install volatility3 ->Create An environment -> open the dump image

Hint: volatility3 is a tool used for scanning dump images linux or windows dumps
It shows some details like size, hashes, Windows, Build Number, Kernel Base, DTB, Architecture (x64/x86) , Processes with PID and PPID

```
{volenv}-(root@kali)-[/home/kali/volatility3]
#python3 vol.py -f /home/kali/Desktop/269-lockdown/memdump.memwindows.info

Volatility 3 Framework 2.28.2
Progress: 100.00 PDB scanning finished
Variable Value

Kernel Base 0xf80079213000
DTB 0x1aa000
Symbols file:///home/kali/volatility3/volatility3/symbols/windows/ntkrnlmp.pdb/EF9A48AFA50FF07C616585BB01919536-1.json.xz
Is64Bit True
IsPAE False
layer_name 0 WindowsIntel32e
memory_layer 1 FileLayer
KdVersionBlock 0xf80079613f10
Major/Minor 15.17763
MachineType 34404
KeNumberProcessors 4
SystemTime 2024-09-10 06:14:13+00:00
NtSystemRoot C:\Windows
NtProductType NtProductServer
NtMajorVersion 10
NtMinorVersion 0
PE MajorOperatingSystemVersion 10
PE MinorOperatingSystemVersion 0
PE Machine 34404
PE TimeDateStamp Sun Nov 10 07:20:39 2075

{volenv}-(root@kali)-[/home/kali/volatility3]
#|
```

Q6: Your memory snapshot captures the system's kernel in situ, providing vital context for the breach. What is the kernel base address in the dump?

```
(volenv)-(root@kali)-[/home/kali/volatility]
# python3 vol.py -f /home/kali/Desktop/269-lockdown/memdump.memwindows.info

Volatility 3 Framework 2.28.2
Progress: 100.00 PDB scanning finished
Variable Value

Kernel Base 0xf80079213000
DTB 0x1aa000
Symbols file:///home/kali/volatility3/volatility3/symbols/windows/ntkrnlmp.pdb/EF9A48AFA50FF07C616585BB01919536-1.json.xz
Is64Bit True
IsPAE False
layer_name 0 WindowsIntel32e
memory_layer 1 FileLayer
KdVersionBlock 0xf80079613f10
Major/Minor 15.17763
MachineType 34404
KeNumberProcessors 4
SystemTime 2024-09-10 06:14:13+00:00
NtSystemRoot C:\Windows
NtProductType NtProductServer
NtMajorVersion 10
NtMinorVersion 0
PE MajorOperatingSystemVersion 10
PE MinorOperatingSystemVersion 0
PE Machine 34404
PE TimeDateStamp Sun Nov 10 07:20:39 2075

(volenv)-(root@kali)-[/home/kali/volatility]
#
```

ANS: 0xf80079213000

Q7: A trusted service launches an unfamiliar executable residing outside the usual IIS stack, signalling a persistence implant. What is the final full on-disk path of that executable?

We said once that we got sample malware from the affected server

We are going to find out its information

If we search with windows.ps-tree or windows.pslist we can find the exe malware file and its information

```
(volenv)-(root@kali)-[/home/kali/volatility]
# python3 vol.py -f /home/kali/Desktop/269-lockdown/memdump.memwindows.ps-tree

Volatility 3 Framework 2.28.2
Progress: 100.00 PDB scanning finished
PID PPID ImageFileName Offset(V) Threads Handles SessionId Wow64 CreateTime ExitTime Audit Cmd Path
*** 900 4332 updatenow.exe 0xce0657ddb1c0 3 - 0 True 2024-09-10 06:08:23.000000 UTC N/A \Device\HarddiskVolume1\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup\updatenow.exe "C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup\updatenow.exe" C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup\updatenow.exe
** 792 628 svchost.exe 0xce0654f69080 14 - 0 False 2024-09-10 05:29:12.000000 UTC N/A \Device\HarddiskVolume1\Windows\System32\svchost.exe C:\Windows\system32\svchost.exe -k
DcomLaunch -p C:\Windows\system32\svchost.exe
```

Its name is updatenow.exe

So, the path is

ANS: C:\ProgramData\Microsoft\Windows\StartMenu\Programs\Startup\updatenow.exe

Q8: The reverse shell's outbound traffic is handled by a built-in Windows process that also spawns the implanted executable. What is the name of this process, and what PID does it run under?

He said the process is built in windows processes so we will keep With the same command, we can find an app called w3wp.exe its PID is 4332

```
*** 4332 2452 w3wp.exe 0xce06574ca080 0 - 0 False 2024-09-10 05:44:45.000000 UTC 2024-09-10 06:10:48.000000 UTC \Device\HarddiskVolume1\Windows\System32\inetrv\w3wp.exe
**** 900 4332 updatenow.exe 0xce0657ddb1c0 3 - 0 True 2024-09-10 06:08:23.000000 UTC N/A \Device\HarddiskVolume1\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup\updatenow.exe "C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup\updatenow.exe" C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup\updatenow.exe
```

Malware Sample Analysis

Tools: Strings or Detect It Easy (DIE), Virus Total site

Q9: Static inspection reveals the binary has been packed to hinder analysis. Which packer was used to obfuscate it?

Go to the path of updatenow.exe

- 1- sudo apt update
- 2- sudo apt install detect-it-easy

Then -> diec "/home/kali/Desktop/269-lockdown/updatenow.exe"

We can see the Packer : UPX

```
(kali) kali-[~]
$ diec "/home/kali/Desktop/269-lockdown/updatenow.exe"
PE32
Format: Autolt(3.XX)
Linker: Microsoft Linker(12.00.40629)
Compiler: Microsoft Visual C/C++(18.00.40629)[POGO_O_CPP]
Tool: Microsoft Visual Studio(2013)
Packer: UPX(3.91)[NRV, brute]

(kali) kali-[~]
$
```

ANS: UPX

Q10: Threat-intel analysis shows the malware beaconing to its command-and-control host. Which fully qualified domain name (FQDN) does it contact?

Now we got to open [VirusTotal](#) -> Upload The file -> Section Behavior ->scroll down -> you will find the ip in DNS Resolver



ANS: cp8nl.hyperhost.ua

Q11: Open-source intel associates that hash with a well-known commodity RAT. To which malware family does the sample belong?

From VirusTotal -> Section Community ->scroll down -> you will find Family

We can find : Family.AGENTTESLA



ANS: AGENTTESLA